Packet Tracer - Back Up Configuration Files

Completed Lab Report with Screenshots

Topology

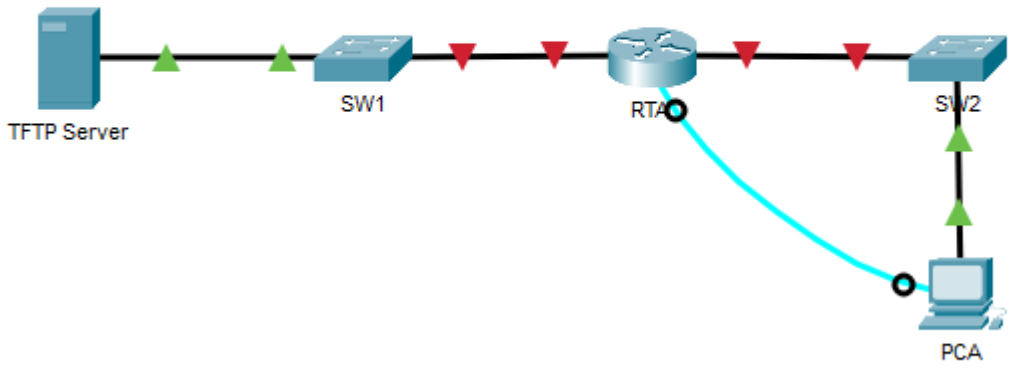


Figure 1 — TFTP Server, SW1, RTA (router), SW2, and PCA connected as specified.

Part 1: Establish Connectivity to the TFTP Server

a. Accessed the RTA command line via PCA > Desktop > Terminal (console connection).

b. Configured and activated interface GigabitEthernet0/0 on RTA with the address matching the TFTP Server's default gateway:

```
Router(config)# interface g0/0
Router(config-if)# ip address 172.16.1.1 255.255.255.0
Router(config-if)# no shutdown
```

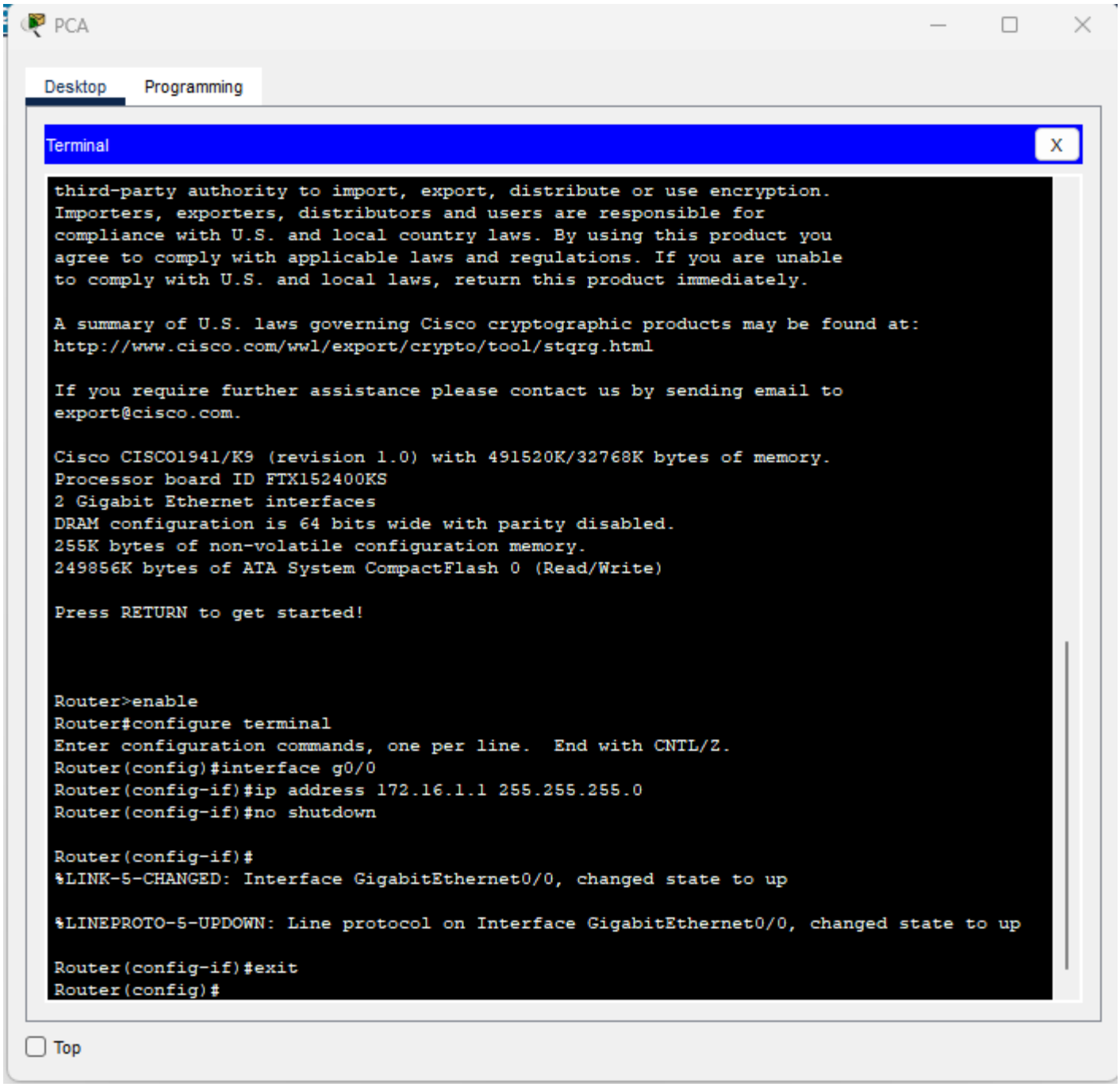


Figure 2 — G0/0 configured; link and line protocol changed to "up".

c. Tested connectivity to the TFTP Server:

```
Router# ping 172.16.1.2
```

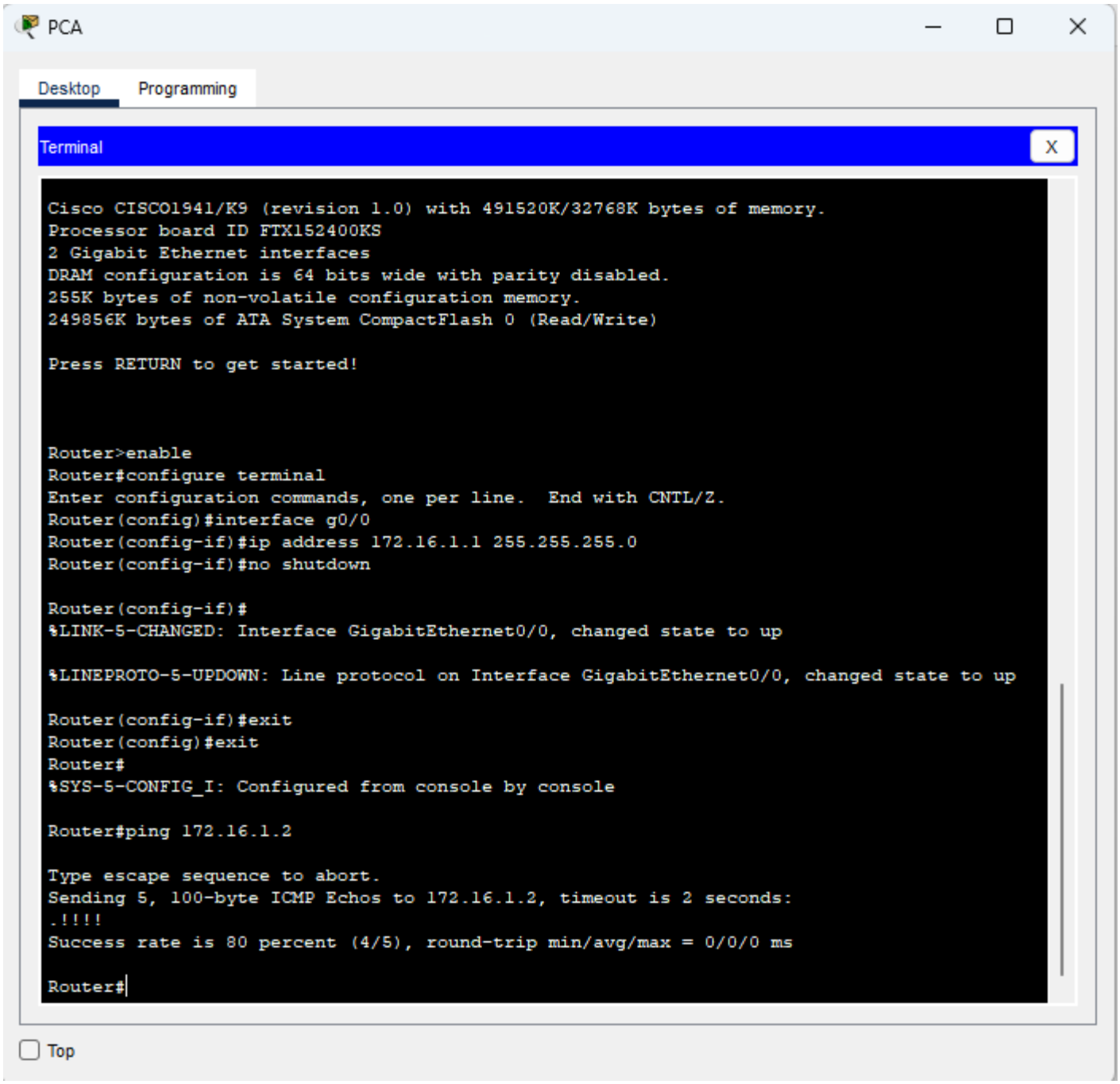


Figure 3 — Ping success rate 80% (4/5); first echo timed out during ARP resolution, remaining four succeeded, confirming connectivity.

Part 2: Transfer the Configuration File from the TFTP Server

a. Restored the saved configuration from the TFTP Server:

```
Router# copy tftp running-config
Address or name of remote host []? 172.16.1.2
Source filename []? RTA-confg
Destination filename [running-config]? <cr>
```

```
Router#copy tftp running-config
Address or name of remote host []? 172.16.1.2
Source filename []? RTA-confg
Destination filename [running-config]?

Accessing tftp://172.16.1.2/RTA-confg...
Loading RTA-confg from 172.16.1.2: !
[OK - 785 bytes]

785 bytes copied in 0 secs
RTA#
%SYS-5-CONFIG_I: Configured from console by console

RTA#
```

Figure 4 — 785 bytes copied successfully; prompt changed from Router# to RTA#.

b. Displayed the current configuration with show running-config.

Question: *What changes were made?*

Answer: The configuration stored on the TFTP Server was loaded into the router and the hostname of the router changed from "Router" to "RTA" (confirmed by the prompt change from Router# to RTA#, and the %SYS-5-CONFIG_I message).

c. Displayed the interface status with show ip interface brief:

```
RTA#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     172.16.1.1      YES manual up                    up
GigabitEthernet0/1     172.31.1.1      YES manual administratively down down
Vlan1                  unassigned      YES unset  administratively down down
RTA#
```

Figure 5 — G0/0 is up/up; G0/1 is administratively down/down; Vlan1 is administratively down.

Question: *Are all interfaces active?*

Answer: No. GigabitEthernet0/0 is up, but GigabitEthernet0/1 is administratively down (Vlan1 is also administratively down, which is normal for a router with no SVI configured).

d. Corrected the interface problem by enabling G0/1, then re-tested connectivity between PCA and the TFTP Server:

```
RTA(config)# interface g0/1
RTA(config-if)# no shutdown
...
RTA# ping 172.16.1.2
```

```
RTA#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RTA(config)#interface g0/1
RTA(config-if)#no shutdown

RTA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

RTA(config-if)#exit
RTA(config)#exit
RTA#
%SYS-5-CONFIG_I: Configured from console by console

RTA#ping 172.16.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

RTA#
```

Figure 6 — G0/1 changed state to up/up; ping to the TFTP Server now succeeds at 100% (5/5).

Part 3: Back Up Configuration and IOS to the TFTP Server

a-b. Changed the hostname to RTA-1 and saved the running configuration to NVRAM:

```
RTA(config)# hostname RTA-1
RTA-1(config)# exit
RTA-1# copy running-config startup-config
Destination filename [startup-config]? <cr>
```

```
RTA-1#copy running-config startup-config
Destination filename [startup-config]?
```

Figure 7 — Hostname changed to RTA-1; startup-config save confirmed with [OK].

c. Backed up the running configuration to the TFTP Server:

```
RTA-1# copy running-config tftp:
Address or name of remote host []? 172.16.1.2
Destination filename [RTA-1-confg]? <cr>
```

```
RTA#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
RTA(config)#hostname RTA-1
RTA-1(config)#exit
RTA-1#
%SYS-5-CONFIG_I: Configured from console by console

RTA-1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
RTA-1#copy running-config tftp:
Address or name of remote host []? 172.16.1.2
Destination filename [RTA-1-confg]?

Writing running-config...!!
[OK - 849 bytes]

849 bytes copied in 0 secs
RTA-1#
```

Figure 8 — Full sequence: hostname change, NVRAM save ([OK]), and 849 bytes copied to the TFTP Server as RTA-1-confg.

d. Displayed the files in flash memory:

```
RTA-1# show flash
```

```
RTA-1#show flash

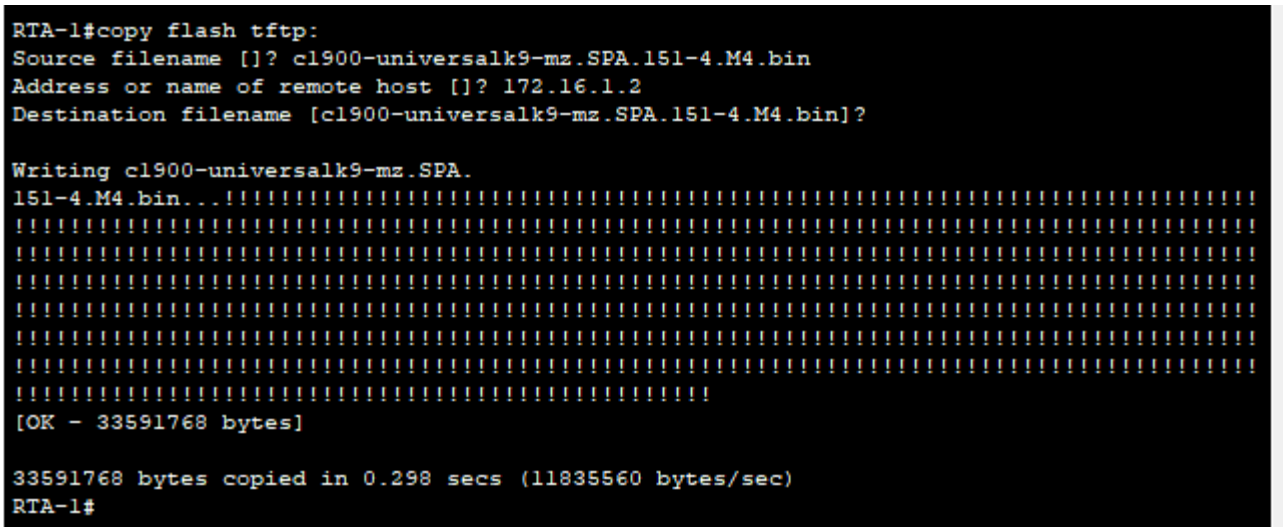
System flash directory:
File  Length   Name/status
  3   33591768 c1900-universalk9-mz.SPA.151-4.M4.bin
  2   28282    sigdef-category.xml
  1   227537   sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)


RTA-1#
```

Figure 9 — Flash directory listing showing the IOS image c1900-universalk9-mz.SPA.151-4.M4.bin (33,591,768 bytes).

e. Backed up the IOS image in flash to the TFTP Server:

```
RTA-1# copy flash tftp:
Source filename []? c1900-universalk9-mz.SPA.151-4.M4.bin
Address or name of remote host []? 172.16.1.2
Destination filename [c1900-universalk9-mz.SPA.151-4.M4.bin]? <cr>
```


RTA-1#copy flash tftp:
Source filename []? c1900-universalk9-mz.SPA.151-4.M4.bin
Address or name of remote host []? 172.16.1.2
Destination filename [c1900-universalk9-mz.SPA.151-4.M4.bin]?
Writing c1900-universalk9-mz.SPA.
151-4.M4.bin...!!!

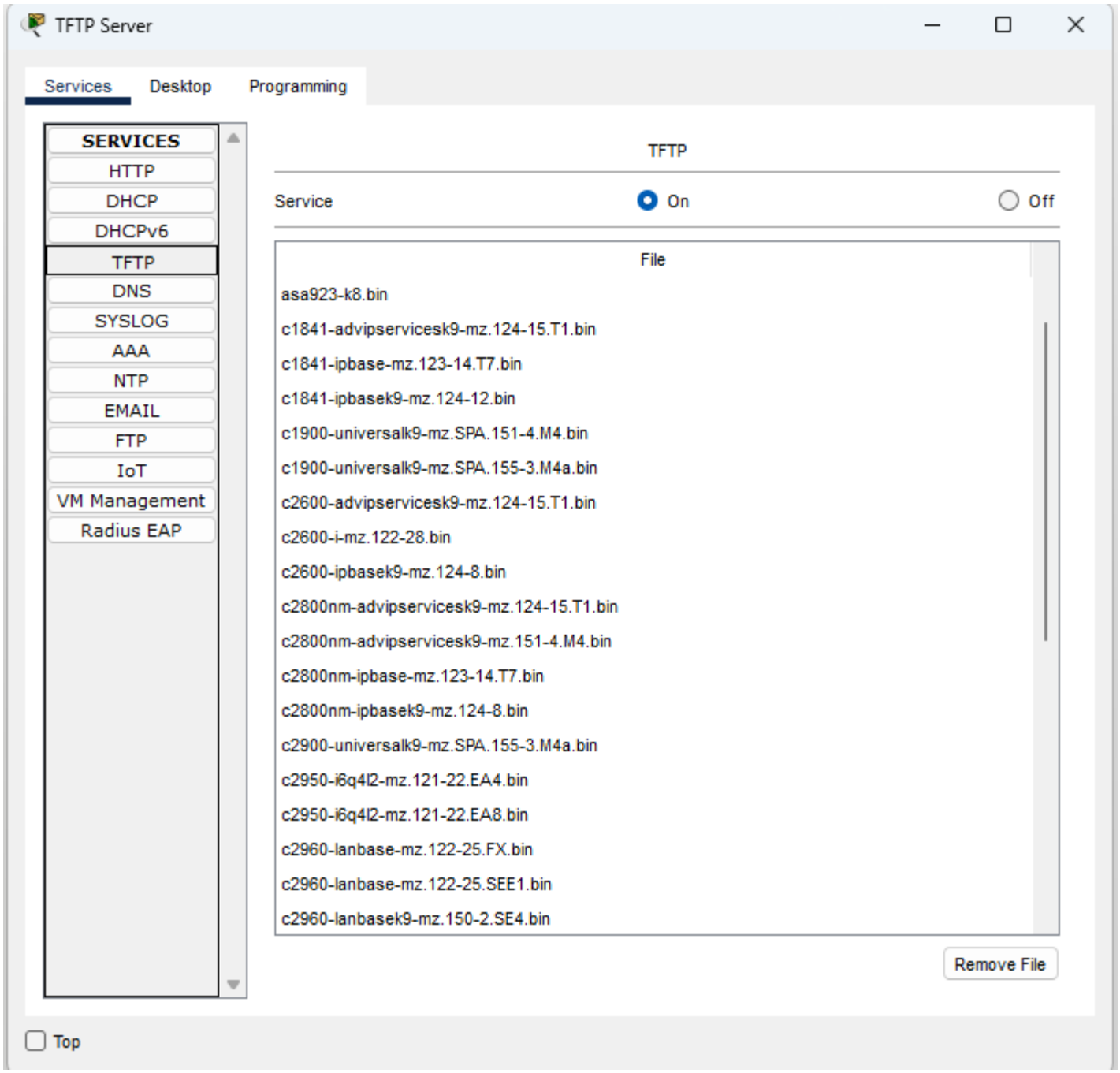


Figure 11 — TFTP Server file list, confirming c1900-universalk9-mz.SPA.151-4.M4.bin is present.

Question: *Has the IOS file c1900-universalk9-mz.SPA.151-4.M4.bin been copied to the TFTP Server?*

Answer: Yes, the c1900-universalk9-mz.SPA.151-4.M4.bin file is listed among the files on the TFTP server.

End of Lab — All Parts (1, 2, and 3) completed successfully.